

Communication Efficient Gradient Coding: A Survey

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Chapter 1

Implementation

In this chapter, we will discuss our own implementation of a toy example of a gradient coding scheme. We will implement a simple gradient coding scheme based on the idea of polynomial interpolation as described in the reference paper. We will use Python for our implementation.

1.1 Setup

We have made an assumption in our implementation.

$$n \text{ (Number of workers)} = k \text{ (Number of datasets)}$$

This is consistent with the reference paper, which also makes a similar assumption for simplicity.

We test the correctness of our implementation by trying out multiple test cases mentioned below:

Test Case	n = k	d	s	m	l
1	5	3	1	2	4
2	6	3	1	2	6
3	8	4	2	2	8
4	10	5	3	2	20
5	10	4	2	2	20
6	10	3	1	2	20
7	12	6	3	3	12
8	15	5	2	3	15

Table 1.1: Test cases for our implementation

n = number of workers, k = number of datasets, d = number of datasets assigned to each worker, s = number of stragglers tolerated, m = communication reduction factor, l = dimension of the gradient vector.